Face rings

Let Δ be a simplicial complex with vertex set $[n]$. Fix a field $\mathbb{k}$. The face ring of Δ is

$$\mathbb{k}[\Delta] := \mathbb{k}[x_1, \ldots, x_n] / I_\Delta$$

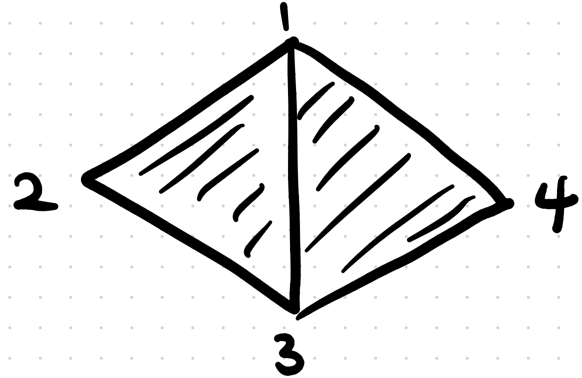
where I_Δ is the ideal of $\mathbb{k}[x_1, \ldots, x_n]$ generated by all elements of the form

$$x_{i_1} x_{i_2} \cdots x_{i_k}, \quad \{i_1, \ldots, i_k\} \notin \Delta .$$

I_Δ is the "non-face ideal"

$$x^S = \prod_{i \in S} x_i$$

$$I_\Delta = \langle x^S : S \notin \Delta \rangle$$

$\Delta =$ 

$$\Bbbk[\Delta] = \Bbbk[x_1, x_2, x_3, x_4] / I_\Delta$$

Every non-face of Δ contains $\{2, 4\}$.

so $I_\Delta = \langle x_2 x_4 \rangle$

$$\Bbbk[\Delta] = \Bbbk[x_1, x_2, x_3, x_4] / \langle x_2 x_4 \rangle$$

A **$\mathbb{k}$-algebra** is a commutative ring A along with a $\mathbb{k}$-vector space structure on A compatible with the ring structure. A $\mathbb{k}$-algebra is **graded** if as a vector space we have

$$A = A_0 \oplus A_1 \oplus \cdots$$

where

(a) $A_0 \cong \mathbb{k}$

(b) $A_i A_j \subseteq A_{i+j}$ for all i, j.

An element $x \in A_i$ is **homogeneous of degree** i.

$$\mathbb{k}[x_1, \dots, x_n] = \bigoplus_{i \geq 0} A_i$$

where A_i is the vector space with basis all monomials of degree i

$$A_0 = \mathbb{k}$$

$$A_1 = \{a_1 x_1 + a_2 x_2 + a_3 x_3 + a_n x_n\}$$

$$\vdots$$

A **graded ideal** of of graded $\mathbb{k}$-algebra

$A = \bigoplus_{i \geq 0} A_i$ is a vector subspace of the form

$I = \bigoplus_{i \geq 0} I_i$, such that $A_i I_j \subseteq I_{i+j}$.

$I_i \subseteq A_i$

Example: If I is generated by finitely many homogeneous elements, then it is graded.

Given a graded ideal $I = \bigoplus_{i \geq 0} I_i$ of $A = \bigoplus_{i \geq 0} A_i$, we have

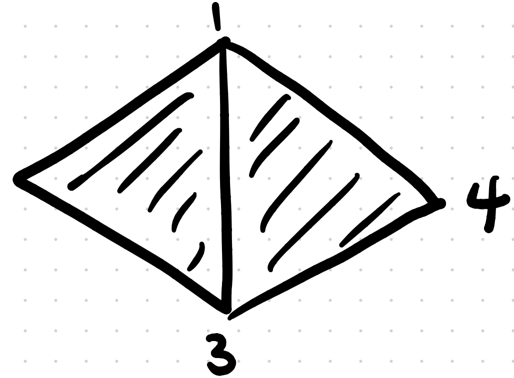
$$A/I \cong \bigoplus_{i \geq 0} (A_i / I_i)$$

with multiplication given by

$$\begin{array}{ccc} \bar{f} \in A_i/I_i & \bar{g} \in A_j/I_j & \\ \uparrow & \uparrow & \bar{f}\bar{g} = fg \pmod{I_{i+j}} \\ f \in A_i & g \in A_j & \end{array}$$

Let $A = \bigoplus_{i \geq 0} A_i$ be a graded $\mathbb{k}$-algebra. The **Hilbert series** of A is the formal power series

$$H(A,t) = \sum_{i=0}^{\infty} (\dim_{\mathbb{k}} A_i)\, t^i$$

$\Delta =$ 

$A = \mathbb{k}[\Delta]$

$A_0 = \mathbb{k}$ $\qquad 1$

$A_1 = \mathrm{span}(x_1, x_2, x_3, x_4)$ $\qquad 4t$

$A_2 = \mathrm{span}(x_1^2, x_2^2, x_3^2, x_4^2, x_1x_2, x_1x_3, x_1x_4, x_2x_3, x_3x_4)$ $\qquad 9t^2$

$$A_3 = \mathrm{span}\big(\underset{\substack{\uparrow\\ 4}}{x_i^3},\ \underset{\substack{\uparrow\\ 10}}{x_i^2 x_j},\ x_1x_2x_3,\ x_1x_2x_4\big) \qquad 16t^3$$

$$A_4 = \mathrm{span}\big(\underset{\substack{\uparrow\\ 4}}{x_i^4},\ \underset{\substack{\uparrow\\ 10}}{x_i^3 x_j},\ \underset{\substack{\uparrow\\ 5}}{x_i^2 x_j^2},\ \underset{\substack{\uparrow\\ 6}}{x_i^2 x_j x_k}\big) \qquad 25t^4$$

Prop: Let Δ be a simplicial complex of dimension d with f-vector $(f_{-1}, f_0, \ldots, f_d)$. Then

$$H(\Bbbk[\Delta], t) = \sum_{i=0}^{d+1} f_{i-1} \frac{t^i}{(1-t)^i}$$

Proof: For each F, $\dim F = i-1$, consider all monomials which have F as its support. The contribution of all such monomials to the Hilbert series is

$$t^i + i t^{i+1} + \binom{i+1}{2} t^{i+2} + \binom{i+2}{3} t^{i+3} + \cdots$$

$$= \frac{t^i}{(1-t)^i}$$

Prop:

$$H(\Bbbk[\Delta], t) = \frac{h_0 + h_1 t + \cdots + h_{d+1} t^{d+1}}{(1-t)^{d+1}}$$

$$\left(\sum_{i=0}^{d+1} f_{i-1} (x-1)^{d+1-i} = \sum_{i=0}^{d+1} h_i x^{d+1-i} \right)$$

Proof:

$$H(\Bbbk[\Delta], t) = \sum_{i=0}^{d+1} f_{i-1} \frac{t^i}{(1-t)^i}$$

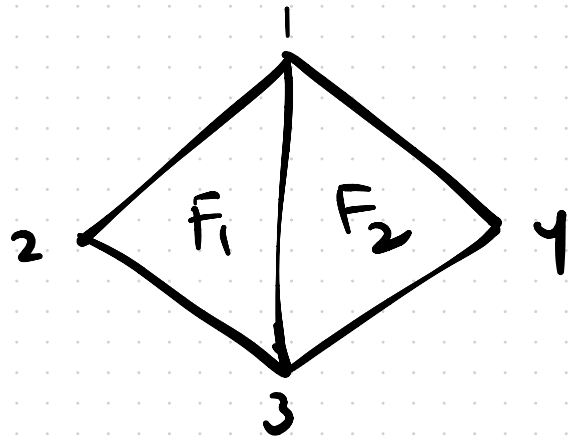
$$= \frac{t^{d+1}}{(1-t)^{d+1}} \sum_{i=0}^{d+1} f_{i-1} \frac{(1-t)^{d+1-i}}{t^{d+1-i}}$$

$$= \frac{t^{d+1}}{(1-t)^{d+1}} \sum_{i=0}^{d+1} f_{i-1} \left(\frac{1}{t} - 1\right)^{d+1-i}$$

$$= \frac{t^{d+1}}{(1-t)^{d+1}} \sum_{i=0}^{d+1} h_i \left(\frac{1}{t}\right)^{d+1-i}$$

$$= \frac{1}{(1-t)^{d+1}} \sum_{i=0}^{d+1} h_i t^i$$

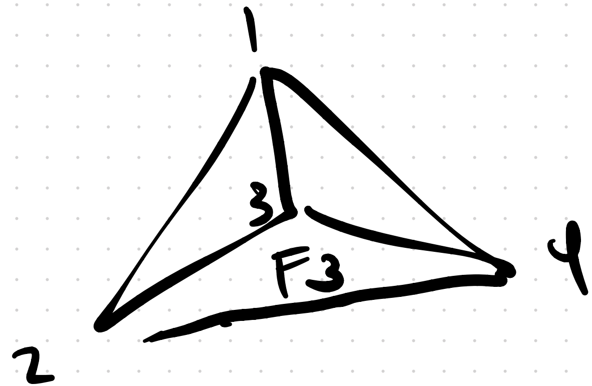
Suppose Δ is shellable.



$$\Bbbk[F_1] = \Bbbk[x_1, x_2 x_3] \qquad H(\Bbbk[F_1], t) = \frac{1}{(1-t)^3}$$

$$\Bbbk[F_1 \cup F_2] = \Bbbk[x_1, x_2, x_3] \oplus x_4 \Bbbk[x_1, x_2, x_4]$$

$$H(\Bbbk[F_1 \cup F_2], t) = \frac{1}{(1-t)^3} + \frac{t}{(1-t)^3}$$



$$\Bbbk[F_1 \cup F_2 \cup F_3] = \Bbbk[F_1 \cup F_2] \oplus x_2 x_4 \Bbbk[x_2, x_3, x_4]$$

$$H(\qquad) = \frac{1}{(1-t)^3} + \frac{t}{(1-t)^3} + \frac{t^2}{(1-t)^3}$$

Prop: If Δ is shellable, then

$$\Bbbk[\Delta] = \bigoplus_{F \text{ facet}} x^{F'} \Bbbk[x_i : i \in F]$$

where F' is the minimal face added to Δ when F is added in the shelling.